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PAM

MIDI to CV Converter

Mono · 1V/oct · 12-bit DAC · Gate + S-Trig

FEATURES

- 12-bit MCP4725 DAC — accurate 1V/oct pitch CV
- Gate output — compatible with most analogue synths
- S-Trig output — onboard jumper selects VTRIG or STRIG
- MIDI channel selection — omni or specific channel, remembered at power-up
- Root note setting — match any keyboard octave convention
- Transpose — shift pitch +/-12 semitones (resets to 0 at power-up)
- On-screen calibration — trim with DVM via CV jack lead
- Portamento/glide — smooth pitch slides between notes
- Note range filter — 7 presets for keyboard splits
- MIDI activity indicator — blue dot flashes on MIDI receive
- EEPROM storage — channel, root, glide, range and calibration survive power cycles
- Colour TFT display — live note, gate, CV readout



Made in Belfast, Northern Ireland

User Guide v3.3 · Alan G Watkins 2026

WHAT IS PAM?

PAM converts MIDI note messages into analogue control voltages (CV) and gate signals, allowing you to play analogue synthesisers from any MIDI keyboard, sequencer or DAW. It follows the 1 volt per octave standard used by most analogue and modular synths.

CONNECTIONS

JACK / CONNECTOR	FUNCTION	NOTES
MIDI IN (DIN5)	MIDI input	Standard 5-pin DIN MIDI
CV>	Pitch CV output	0-5V, 1V/oct
GATE>	Gate/Trig output	Selected by onboard jumper
DC 7-12V	Power input	Centre positive, regulated

GATE TYPE JUMPER (VTRIG / STRIG)

PAM has a 3-pin jumper on the board labelled GATE TYPE to select the gate signal type. The centre pin (pin 2) is common. Fit the jumper to select:

JUMPER POSITION	SIGNAL TYPE	USE WITH
Pins 1-2 (right) DEFAULT	VTRIG — active high 5V gate	Most modern synths
Pins 2-3 (left)	STRIG — active low trigger	Moog, early Korg

PAM is shipped with the jumper in the VTRIG (pins 1-2) position.

MIDI CHANNEL

PAM can respond to all MIDI channels (OMNI) or a single specific channel.
Press button A to cycle: OMNI -> CH1 -> CH2 -> ... -> CH16 -> OMNI
The current channel is shown in the top right of the display.
The selected channel is saved to EEPROM and restored at power-up.

ROOT NOTE SETTING

Different keyboards use different MIDI note numbers for their lowest C. Press button C to cycle the root note convention to match your keyboard:

C1 (MIDI 24)	Most modern keyboards and DAWs — default setting
C2 (MIDI 36)	Some keyboards label their lowest C as C1 but send MIDI 36
C3 (MIDI 48)	Older keyboards, some Yamaha models
C4 (MIDI 60)	Some bass synths and older instruments

The setting is saved to EEPROM and restored on power-up.

TRANSPOSE

Press button B to shift pitch up one semitone per press (-12 to +12). **Transpose is not saved: PAM always starts with transpose at 0**, so a forgotten offset can never leave your synth out of tune after a power cycle.

MAIN DISPLAY

NOTE	Current MIDI note name and number. Shows --- when no note held.
dot	Blue dot top-right of header flashes on MIDI activity.
GATE	GREEN = gate high (note held) GREY = gate low (no note)
CV	Current DAC output voltage and raw count. Yellow bar shows level.
TRAN	Current transpose. Shows none at zero.
ROOT	Keyboard root: C1 / C2 / C3 / C4. Orange when not default.
GLIDE	Current portamento setting. MAGENTA when active, grey when off.
STRIG	MAGENTA = S-Trig active GREY = S-Trig idle

BUTTONS — NORMAL MODE

A	Cycle MIDI channel (OMNI -> 1 -> 2 -> ... -> 16 -> OMNI). Saved automatically.
A hold	PANIC — all notes off, DAC returns to C1 voltage
B	Transpose — shift pitch up one semitone per press (-12 to +12). Resets to 0 at power-up.
B hold	Open help screen
C	Cycle root note convention (C1 / C2 / C3 / C4)
C hold	Open settings (portamento + note range + calibration)

BUTTONS — SETTINGS SCREEN

A	Move cursor between menu items
B	Change value of selected item
C	Enter / confirm (calibrate only)
C hold	Exit settings

SETTINGS — NOTE RANGE PRESETS

FULL	All notes 0-127 — no filtering (default)
LOW OCT	C1 to B2 — lower two octaves
MID LOW	C2 to B3 — mid-bass range
MID	C3 to B4 — middle keyboard range
MID HIGH	C4 to B5 — upper mid range
HIGH	C5 and above — upper keyboard
BASS	C1 to B1 — single bass octave only

BUTTONS — CALIBRATION MODE

A	Coarse up +10 counts
A hold	Coarse down -10 counts
B	Fine up +1 count
B hold	Fine down -1 count
C	Advance to next step / Save and exit on final step
C hold	Exit calibration without saving

CALIBRATION PROCEDURE

Device is factory calibrated but there may be occasions you want to compensate for tolerance in your 5V power supply. Plug a jack lead into the CV> socket and touch your DVM probes to the tip and sleeve of the free end.

STEP 1 Root voltage — C1 = 1.000V

- Hold C button to enter calibration mode.
- The DAC outputs C1. Gate goes high to sound the synth.
- Use A / A-hold for coarse adjustment (+/-10 counts).
- Use B / B-hold for fine adjustment (+/-1 count).
- Adjust until DVM reads exactly 1.000V.
- Press C to advance to step 2.

STEP 2 Scale factor — C2 = 2.000V

- The DAC now outputs C2 (one octave up from C1).
- Touch DVM probes to jack tip and sleeve again.
- Adjust using the same A and B buttons.
- Adjust until DVM reads exactly 2.000V.
- Press C to save calibration and exit.

CALIBRATION RESET screen (new in v3.3). If PAM finds that its stored calibration is missing or outside the valid range at power-up, it loads the factory defaults and shows a red **CALIBRATION RESET** message during start-up. Hold C to open Settings, choose CALIBRATE and repeat Steps 1 and 2. If the message never appears, your calibration was loaded normally.

QUICK SETUP GUIDE

1. Set jumper to VTRIG (pins 1-2) or STRIG (pins 2-3) as needed.
2. Connect MIDI IN to your keyboard or sequencer.
3. Connect CV> to your synth pitch CV input.
4. Connect GATE> to your synth gate/trigger input.
5. Power on PAM. CV output stabilises at 1V (C1) immediately.
6. Press C to cycle root note until notes play at correct octave.
7. Press B to transpose if fine pitch offset is needed.
8. Hold C to enter calibration and trim voltages with DVM via CV jack.

SPECIFICATIONS

Processor	ESP32-WROOM-32E
DAC	MCP4725 / ZC576300 12-bit I2C
CV output range	0 - 5V (1V/oct)
CV resolution	12-bit (4096 steps)
Gate output	VTRIG 5V active high or STRIG active low (jumper select)
MIDI input	Standard DIN5, 31250 baud, optoisolated
Display	ST7735S 1.8" colour TFT 160x128
Power input	7 - 12V DC, centre positive
Note priority	Last note, 8 note stack
Root note options	C1 (MIDI 24) / C2 (MIDI 36) / C3 (MIDI 48) / C4 (MIDI 60)
Transpose range	-12 to +12 semitones (resets to 0 at power-up)
MIDI channels	Omni or single channel 1-16 (saved)
Settings storage	EEPROM — channel, root note, glide, note range and calibration survive power cycles
Firmware	v3.3

Specifications subject to change. Visit www.mindburner.co.uk for latest information.